Highlights

Mostly ranking, partly threshold: decomposing and repairing the cross-corpus generalisation gap of misinformation detectors

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- Ten audited corpora, six detectors, five seeds: train on one, test on all.
- Most of the cross-corpus F1 gap is lost ranking; the rest is a mis-set threshold.
- EM label-shift correction worsens the threshold: calibration does not transfer.
- Flagging the source base rate of new items recovers most threshold loss, label-free.
- Bengali clickbait transfers across publishers and languages; veracity does not.

Mostly ranking, partly threshold: decomposing and repairing the cross-corpus generalisation gap of misinformation detectors

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Abstract

Fake-news detectors score near-perfectly on their training corpus and much worse elsewhere, but cross-domain studies report this with accuracy or F1, which conflates two failures with different remedies: ranking new items badly, and ranking them well but thresholding them wrongly. We separate the two. Ten public corpora in English and Bengali (articles, headlines, claims, tweets) are audited for label direction and near-duplicates, and six detectors are evaluated train-on-one, test-on-all with five seeds, bootstrap intervals and permutation controls. The audit finds that the two most used article corpora code labels in opposite directions, which turns moderate transfer into an apparent below-chance collapse, as well as label-conflicting duplicates and tweets shared across rumour corpora. Mean ROC-AUC falls from 0.914 in-domain to 0.567 across corpora. Decomposing the lost macro-F1 attributes 70% to a ranking deficit and 30% to the operating point. The standard label-shift correction, EM prior adaptation, enlarges the operating-point loss because calibration does not transfer (expected calibration error 0.046 in-domain, 0.328 across corpora), whereas flagging the source's base rate of target items recovers 81% of it without target labels. Twenty-two surface statistics, publisher identity or 64 training examples recover much of the in-domain signal, and frozen RoBERTa features transfer no better than TF-IDF. Pooling six training corpora raises held-out AUC from 0.567 to 0.660, while removing corpus identity from the representation lowers it, because corpus and label are entangled. Bengali clickbait detection, in contrast, survives unseen publishers and languages. We release code, per-item scores and a reporting protocol.

Keywords: misinformation detection, fake news, cross-corpus generalisation, dataset audit, label shift, shortcut learning, Bengali

1. Introduction

False stories travel farther and faster than true ones on social platforms (Vosoughi et al., 2018), and the cost of that asymmetry is borne by public health, elections and trust in institutions (Lazer et al., 2018).

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Automatic detection is one response, and on the standard benchmarks it appears close to solved: content-based classifiers routinely report accuracies above 95% (Shu et al., 2017; Zhou and Zafarani, 2020; Kaliyar et al., 2021). A detector, however, is deployed on data it was not trained on, and a growing body of work shows that these scores do not survive a change of corpus (Pérez-Rosas et al., 2018; Janicka et al., 2019; Suprem and Pu, 2022; Dell'Oglio et al., 2026) and that some of them reflect dataset artefacts rather than veracity (Asr and Taboada, 2019; Thibault et al., 2025; Verma, 2026).

What is still missing is an account of *why* the transfer gap arises in a form that tells a practitioner what to do about it. A detector can fail on a new corpus in two different ways. It can *order* the new items badly, scoring fabricated stories no higher than genuine ones; or it can order them well but place its decision threshold in the wrong place, because the new corpus has a different base rate or because its scores are scaled differently. The first failure needs a better representation; the second only needs a better operating point. Accuracy and F1, the metrics almost every cross-domain study reports, conflate the two.

This paper separates them. We assemble ten public corpora in two languages that cover four text units (articles, headlines, claims and tweets) and three constructs (veracity, rumour and clickbait), audit them for label direction and for within- and cross-corpus duplication, and evaluate six detectors in a train-on-one, test-on-all design with five seeds, bootstrap confidence intervals and permutation controls. We then decompose the macro-F1 that a detector loses on a new corpus into a ranking deficit, which no threshold can recover, and operating-point losses, which the right threshold would, and we evaluate label-free repairs aimed at each part. Our contributions are:

1. **An audited multi-corpus benchmark.** The two most used article corpora code their labels in opposite directions (ISOT: 1 = fake; WELFake: 1 = real). We show that simple signature probes detect such an inversion, which otherwise masquerades as a below-chance collapse of transfer, as it did in our own first pipeline. MinHash deduplication finds label-conflicting duplicates in GossipCop and source tweets shared between PHEME and Twitter15/16 (Section 3).
2. **A transfer gap decomposition.** Across 42 directed corpus pairs, 70% of the macro-F1 gap is a ranking deficit and 30% an operating-point loss (Section 5.2). Mean transfer AUC is 0.567 against 0.914 in-domain; the gap is real but is not the collapse to chance that F1 alone suggests.
3. **A label-free operating-point repair, and why the standard one fails.** Expectation–maximisation prior adaptation, the textbook remedy for label shift, enlarges the operating-point loss (-64% recovered), because calibration does not survive the change of corpus. Transferring the ranking rule instead (flagging the source's base rate of target items) recovers 81% without any target label (Section 5.3).
4. **Evidence on what the detectors learn.** A model of 22 surface statistics and group identity alone recover much of the in-domain signal, 64 examples suffice on the article and tweet corpora, and the

strongest cues are outlet signatures and time expressions; masking, feature deletion and topic balancing show how much of the ranking survives without them (Section 5.5). For the ranking deficit, training on several corpora at once raises held-out AUC from 0.567 to 0.660, whereas making the representation invariant to corpus identity (feature deletion, CORAL, nullspace projection) lowers it, because corpus identity and label are entangled in these benchmarks (Section 5.6).

5. **A low-resource extension.** In Bengali, clickbait detection replicates the published baseline, survives held-out publishers and transfers across languages by translation, which suggests that whether a cue transfers depends on whether it *is* the construct or merely correlates with it (Section 5.7).
6. **A reporting protocol and a full release** of code, seeds, per-cell results and predictions (Section 6).

2. Related work

Cross-domain failure of fake-news detectors. That content-based detectors degrade across domains has been reported since the first multi-domain corpora: models trained on one news domain lose much of their accuracy on another (Pérez-Rosas et al., 2018; Janicka et al., 2019), cross-domain training needs domain-aware architectures (Silva et al., 2021), and fine-tuning on one COVID-19 corpus often hurts performance on related ones (Suprem and Pu, 2022). Bozarth and Budak (2020) showed that conclusions about which model is best depend on the dataset, the evaluation type and the metric. Most recently, Dell'Oglio et al. (2026) benchmarked twelve approaches on ten English corpora and found cross-dataset median F1 close to 0.5 for classical and fine-tuned models, with zero-shot large language models more stable, and Hoy and Koulouri (2025) found stylistic and "social-monetisation" features to transfer better than token representations. These studies establish *that* detectors fail to transfer, almost always through a single threshold-dependent score (accuracy or F1). They do not separate a loss of discrimination from a mis-set decision threshold, which is the distinction this paper is built on.

Shortcuts and dataset artefacts. Shortcut learning, where a model exploits a spurious cue that predicts the label in the training distribution only, is a general failure mode of deep and shallow models alike (Geirhos et al., 2020; Gururangan et al., 2018; McCoy et al., 2019). For misinformation, Horne and Adali (2017) and Schuster et al. (2020) showed how much of the signal is style rather than veracity, and Asr and Taboada (2019) documented topical imbalance across fake-news corpora. Verma (2026) audited the ISOT corpus in depth: subject metadata alone separates the classes perfectly, a newswire tag marks 99.2% of real articles, duplicates contaminate 19.4% of a naive test split, deleting the 1,000 highest-weight unigrams leaves F1 at 0.926, and ISOT-trained models reach only near-chance ranking on LIAR. Thibault et al. (2025) surveyed 75 misinformation datasets and found keyword–label correlations and temporal leakage in several of them, including Twitter15/16, ISOT and BanFakeNews. We build directly on these audits but differ in scope and

in aim: we audit ten corpora jointly, including cross-corpus duplication and label direction, and we use the audit to explain and repair the transfer gap rather than only to document it.

Rumour benchmarks. Rumour-detection benchmarks share the problem. On Twitter15/16 and PHEME, event-level spurious correlations inflate scores (Wu and Hooi, 2022), chronological splits lower them (Mu et al., 2023), and simple text baselines rival propagation-based models (Pelrine et al., 2021). We therefore treat the source posts of these corpora as one more text unit in our matrix and do not re-evaluate propagation models, which these works already cover.

Label shift and invariant representations. Two families of methods from the domain-adaptation literature map onto the two parts of our decomposition. Label-shift adaptation re-estimates the target class prior from unlabelled predictions and adjusts the posteriors (Saerens et al., 2002; Lipton et al., 2018); with calibrated scores, this simple expectation–maximisation procedure is hard to beat (Alexandari et al., 2020). Representation methods instead try to remove domain information: correlation alignment matches feature covariances (Sun et al., 2016), adversarial training removes domain-predictive directions (Ganin et al., 2016), and iterative nullspace projection deletes linearly decodable attributes (Ravfogel et al., 2020). Domain-adaptation theory bounds target error by source error plus a domain divergence that can be estimated with a domain classifier (Ben-David et al., 2010). In misinformation, label shift has been addressed mainly with pseudo-label correction inside neural adaptation pipelines; we are not aware of work that measures how much of the cross-corpus gap is a prior/threshold effect and how much a representation effect, and repairs each with the matching tool.

Large language models. Zero-shot and few-shot large language models are less exposed to dataset-specific shortcuts because they are not trained on the benchmark (Pelrine et al., 2023; Thibault et al., 2025; Dell'Oglio et al., 2026), and they can serve as advisers to smaller detectors (Hu et al., 2024). Supervised detectors nonetheless remain the cheapest and most widely deployed option, and the decomposition we propose applies to any detector that outputs a score, including a prompted language model.

Position of this work. Table 1 contrasts our study with the closest prior audits. The combination we contribute is a two-way, multi-corpus, multi-unit and bilingual matrix; a decomposition that attributes the gap to ranking or to the operating point; label-free repairs targeted at each part; and audit findings (label inversion, cross-corpus duplication) that change measured transfer.

3. Corpora and data audit

3.1. Corpora

We use ten public corpora (Table 2). Seven share the veracity construct and form the main transfer matrix: two article corpora, ISOT (Ahmed et al., 2017, 2018) and WELFake (Verma et al., 2021a); two

Table 1: Position relative to the closest prior audits: S&P = Suprem and Pu (2022); T = Thibault et al. (2025); V = Verma (2026); D = Dell'Oglio et al. (2026) (✓ present, – absent; units: a = article, h = headline, c = claim, t = tweet).

	S&P	T	V	D	**This work**
Corpora (languages)	9 (en)	75 (survey)	2 (en)	10 (en)	10 (en, bn)
Text units	mixed	mixed	a, c	mixed	a, h, c, t
Full train×test transfer matrix	✓	–	–	✓	✓
Within- and cross-corpus near-duplicate control	–	partial	✓	–	✓
Label-direction verification	–	–	–	–	✓
Threshold-free metrics (AUC, AP)	–	–	✓	–	✓
Ranking vs. operating-point decomposition	–	–	partial	–	✓
Seeds, bootstrap CIs, permutation control	–	–	✓	tests only	✓
Label-free repair evaluated	reject opt.	–	–	MERMAID	EM, rank, INLP, CORAL
Low-resource language	–	–	–	–	Bengali

headline corpora from FakeNewsNet, PolitiFact and GossipCop (Shu et al., 2020), for which the public release contains titles and URLs; the claim corpus LIAR (Wang, 2017), obtained through LIAR-PLUS (Alhindi et al., 2018), binarised as {pants-fire, false} versus {mostly-true, true} with the two middle grades dropped; and the source tweets of Twitter15 and Twitter16 (Ma et al., 2017), using their false and true classes. Three further corpora extend the analysis to other constructs and to Bengali: the source tweets of PHEME rumour/non-rumour threads (Zubiaga et al., 2016, 2017); an English clickbait headline corpus (Chakraborty et al., 2016); and BanglaBait (Mahtab et al., 2023), 15,056 labelled Bengali news articles with machine-translated English titles. Labels differ in what they assert: LIAR rates individual claims, FakeNewsNet and Twitter15/16 rate a story through a fact-check, and ISOT and WELFake label articles largely by the outlet they came from. We keep these distinctions explicit and draw the strongest conclusions from four *same-task pairs* that hold unit and construct fixed: ISOT↔WELFake (articles), PolitiFact↔GossipCop (headlines, politics vs. entertainment), Twitter15↔Twitter16 (tweets, two collection windows), and LIAR↔PolitiFact (claims vs. headlines from the same fact-checker). Every label is mapped to 1 = fake, false, rumour or clickbait.

3.2. Audit

Label direction. Corpora disagree on which value denotes "fake", and nothing in a CSV file says which convention it follows. We therefore verify label direction with *signature probes*: strings that only a reliable or only a fabricating outlet produces. Table 3 shows the result for the two article corpora. In ISOT, 20,761 of the 20,768 "(Reuters)" datelines carry label 0 and all 6,098 texts with the fake-site template "Featured

Table 2: Corpora after harmonisation and deduplication. "Exact" and "near" count removed duplicates; "conflict" counts items removed because duplicates disagreed on the label; "pos." is the share labelled fake, false, rumour or clickbait. For LIAR and Twitter15/16, "raw" counts the items in the binary view (LIAR: pants-fire/false vs. mostly-true/true; Twitter: false vs. true).

Corpus	Unit	Construct	Lang.	Raw	Exact	Near	Conflict	Final	Pos.
ISOT	article	veracity	en	38,833	0	79	0	38,749	0.46
WELFake	article	veracity	en	24,530	0	815	6	23,708	0.43
PolitiFact	headline	veracity	en	1,056	71	3	6	976	0.43
GossipCop	headline	veracity	en	22,140	1,174	26	380	20,560	0.23
LIAR	claim	veracity	en	8,090	21	11	3	8,055	0.44
Twitter15	tweet	veracity	en	742	55	14	0	673	0.51
Twitter16	tweet	veracity	en	412	51	3	0	358	0.46
PHEME	tweet	rumour	en	5,802	124	98	25	5,555	0.34
ClickbaitEN	headline	clickbait	en	32,000	2	123	0	31,875	0.50
BanglaBait	headline	clickbait	bn	15,056	850	124	41	14,041	0.34

image via" carry label 1, so 1 = fake. In WELFake the opposite holds: all 6,223 titles ending in "- The New York Times" carry label 1 and all 125 "Featured image via" texts carry label 0, in agreement with the official documentation (0 = fake, 1 = real; Verma et al., 2021b). Our own first pipeline had attached the more common convention (1 = fake) to the WELFake file. Read that way, every ISOT↔WELFake score is inverted and ISOT→WELFake looks like a below-chance collapse; the probes exposed the error. A ranking metric below 0.5 on a same-task pair is therefore a symptom to investigate before it is a finding to report.

Duplicates. Within each corpus we removed exact duplicates of the normalised text and then near-duplicates found by MinHash locality-sensitive hashing (Broder, 1997) (128 permutations, Jaccard ≥ 0.8; word 3-grams for articles, character 5-grams otherwise). Groups whose members disagree on the label were dropped entirely. Table 2 reports the counts. Two corpora stand out. GossipCop contains 1,174 redundant exact copies of headlines, and a further 356 rows whose headline appears under *both* labels; BanglaBait contains 850 redundant exact copies of titles and, after deduplication, 14,041 items. The leakage-controlled WELFake copy still contained 815 near-duplicates.

Cross-corpus overlap. Near-duplicates across corpora inflate transfer. ISOT and WELFake share only 18 near-duplicate articles, so the leakage control of the WELFake copy is effective. PHEME, however, shares 58 source tweets with Twitter15 and 128 with Twitter16, because the corpora were collected around overlapping breaking-news events; Twitter15 and Twitter16 share 169 tweet identifiers, all labelled *unverified*. Every transfer test excludes target items that are near-duplicates of the source corpus.

Table 3: Label-direction probes on the raw files: number of items matching each source signature under raw label 0 / raw label 1. A reliable-outlet signature must sit under "real", a fake-site signature under "fake". ISOT codes 1 = fake; the WELFake file codes 1 = real.

Signature	Expected class	ISOT (0 / 1)	WELFake (0 / 1)
title ends '- The New York Times'	real	0 / 0	0 / 6,223
title ends '- Breitbart'	real	0 / 0	1 / 2,338
text contains '(Reuters)'	real	20,761 / 7	8 / 23
text contains 'Featured image via'	fake	0 / 6,098	125 / 0
title contains 'BREAKING'	fake	8 / 614	199 / 19
text contains '21st Century Wire'	fake	0 / 629	5 / 0
text contains 'Activist Post'	fake	0 / 5	21 / 0

4. Methodology

4.1. Evaluation protocol

For every corpus and every seed $s \in \{0, \ldots, 4\}$ we draw a stratified 70/10/20 train/validation/test split; the seed also fixes model initialisation and any subsampling. Training splits are capped at 8,000 items, validation at 1,500 and test at 3,000 (stratified), so that the largest corpora do not dominate run time; every cap is applied identically to every model. Featurisers (vocabularies, IDF weights, SVD bases) are fitted on the source training split only. Hyperparameters (the inverse regularisation strength $C \in \{0.25, 1, 4\}$, or the number of boosting rounds) are selected on the source validation split by ROC-AUC, and the selected model's scores are then Platt-calibrated (Platt, 1999) on the same validation split, so the default operating point $t = 0.5$ is the Bayes decision under the source prior. No hyperparameter, threshold or calibration step ever sees a target label. When a model trained on corpus A is tested on corpus B, test items of B that are near-duplicates of any item of A (Section 3.2) are removed.

Three training regimes are run for every model. *Full* uses the capped training split. *Matched* subsamples every source to $n = 250$ training items, removing the size difference between Twitter16 (250 training items) and the article corpora (8,000). *Permuted* shuffles the source training and validation labels; it is a sanity control whose transfer and in-domain AUC must be close to 0.5.

4.2. Models

We use six detectors that span sparse, dense and pretrained representations, all with a linear or tree-based decision layer so that their behaviour is interpretable and their training is reproducible on a CPU: (i) TF-IDF word 1–2-grams with logistic regression (TFIDF-LR), the primary instrument; (ii) the same features with a linear SVM (TFIDF-SVM); (iii) NB-SVM, logistic regression on naive-Bayes-weighted binary n-grams

(Wang and Manning, 2012); (iv) averaged GloVe 100-d embeddings (Pennington et al., 2014) with logistic regression (GLOVE-LR); (v) a style model, gradient boosting on 22 surface statistics (length, punctuation and casing rates, digit and stop-word ratios, pronoun rates, weekday mentions, "said" rate, dateline, URL, handle and hashtag rates; STYLE-GB), which sees no vocabulary beyond counts of stop words, pronouns, weekday names and "said", and therefore measures how far a corpus is separable by surface form; (vi) RoBERTa-base (Liu et al., 2019) mean-pooled last-layer representations with logistic regression (ROBERTA-LR), computed on a stratified subset of at most 1,500 items per corpus (first 600 characters), where TFIDF-LR is refitted on the identical splits for a paired comparison. All models use scikit-learn (Pedregosa et al., 2011); RoBERTa weights are those packaged in spaCy's `en_core_web_trf` 3.8.0 (Honnibal et al., 2020).

4.3. Metrics

For every (model, regime, source, target, seed) cell we report threshold-free ranking metrics (ROC-AUC, average precision), threshold-dependent metrics at the default operating point (macro-F1, per-class F1, balanced accuracy, Matthews correlation, predicted-positive rate), calibration (expected calibration error, 15 bins), and macro-F1 at four further operating points described next. Because test sets are balanced to different degrees, two reference levels are useful when reading macro-F1: random guessing gives about 0.5, and predicting a single class gives about 0.33. Uncertainty is reported as mean $\pm$ standard deviation over the five seeds and as a pooled seed-bootstrap 95% confidence interval (200 resamples of the target test set per seed, pooled over seeds), which captures both split and test-sampling variability. Paired model comparisons use McNemar's test (McNemar, 1947) per cell and seed; repair comparisons use Wilcoxon signed-rank tests over (target, seed) pairs; families of tests are Holm-corrected (Holm, 1979).

4.4. Transfer gap decomposition

Let $F_{AB}(\tau)$ be the macro-F1 on target B of a model trained on source A, at operating point τ. We consider four operating points on the target scores: the default threshold $\tau_0 = 0.5$; the prior-matched threshold τ_π, which makes the predicted-positive rate equal to the true target prior; and the oracle threshold $\tau^\star = \arg\max_\tau F_{AB}(\tau)$, an upper bound that any thresholding rule could at best reach given the model's ranking of target items. The loss a user experiences when deploying the source model on B, relative to what an in-domain model of B achieves, then decomposes exactly:

$$\underbrace{F_{BB}(\tau_0) - F_{AB}(\tau_0)}_{\text{transfer gap}} = \underbrace{\big[F_{BB}(\tau_0) - F_{AB}(\tau^\star)\big]}_{\text{ranking deficit}} + \underbrace{\big[F_{AB}(\tau^\star) - F_{AB}(\tau_\pi)\big]}_{\text{threshold residual}} + \underbrace{\big[F_{AB}(\tau_\pi) - F_{AB}(\tau_0)\big]}_{\text{prior / calibration loss}}. \tag{1}$$

The first term cannot be recovered by any choice of threshold: it measures how much worse the source model *orders* target items than the target's own model does. The last two terms are *operating-point* losses: they would vanish with the right threshold. Adding $F_{AA}(\tau_0) - F_{BB}(\tau_0)$ gives the gap relative to the source's own in-domain score, which is what a benchmark user is promised; this difficulty term averages to zero over a

full matrix and is reported per cell only. The same decomposition is applied with ROC-AUC, where only the ranking term exists.

4.5. Label-free operating points

Three operating points need no target labels. (a) *EM prior adaptation* (Saerens et al., 2002) re-estimates the target prior from the source-calibrated posteriors by expectation–maximisation, adjusts the posteriors and thresholds them at 0.5; with Platt-calibrated scores this is the bias-corrected recipe recommended by Alexandari et al. (2020). It is optimal under pure label shift. (b) *Rank transfer* sets the threshold so that the predicted-positive rate on the target equals the source prior π_A, i.e. it transfers the model's ranking-based decision rule rather than its score scale. (c) A *2-component Gaussian mixture* on the target logits places the threshold where the two components' posteriors cross. For reference we also report τ_π, which needs one number about the target (its base rate) but no item labels.

4.6. Mechanism probes

To identify what signal carries in-domain performance, we use eight probes, each run with TFIDF-LR unless stated: *surface statistics only* (STYLE-GB above); *group-only*, out-of-fold smoothed target encoding of group identity (publisher domain for PolitiFact, GossipCop and BanglaBait; speaker for LIAR; event for PHEME), i.e. how well the label is predicted by *who* published an item; *learning curves* from 32 training items to the full split; *top-K deletion*, removing the $K \in \{10, \ldots, 3000\}$ highest-$|w|$ features and retraining (cf. Verma, 2026); *masking* of source tags and datelines, of time expressions and numbers, of named entities (spaCy `en_core_web_sm`), or of all three; *topic-balanced training*, where classes are subsampled to equal size inside each of 50 k-means topic clusters, removing topic–label correlation from the training set; *corpus separability*, the proxy $\mathcal{A}$-distance $2(1-2\epsilon)$ between two corpora, with ϵ the cross-validated error of a classifier that tells them apart from text alone (Ben-David et al., 2010); and *feature attribution*, the largest-weight n-grams per class and their linear-SHAP magnitudes $|w_j|\,\mathbb{E}|x_j - \bar{x}_j|$. For the more expensive probes (top-K, masking, vocabulary ablation) training is capped at 4,000 and test at 1,500 items; masking and vocabulary ablation use three seeds.

4.7. Representation repairs

The ranking deficit can only be reduced by a better representation. We test repairs in a leave-one-corpus-out (LOCO) setting: for each held-out veracity corpus, the model is trained on the union of the other six (each capped at 1,500 training and 300 validation items) and evaluated on the held-out test split, with near-duplicates of any training corpus removed. Methods are: *pooled* training; pooled training with sample weights that equalise every (corpus, class) cell (*balanced*); balanced training on texts with source tags, datelines, URLs, handles and time expressions *masked*; balanced training after deleting the 2,000 features

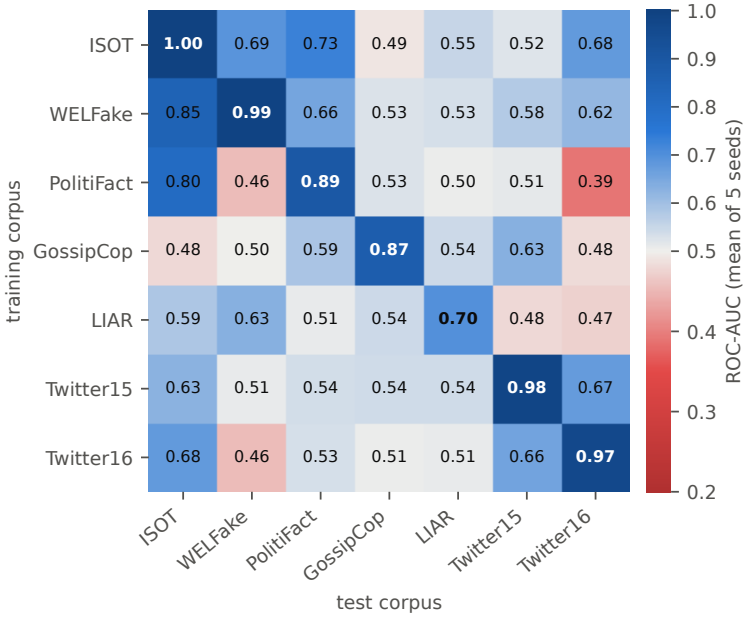


Figure 1: Transfer ROC-AUC of TFIDF-LR (mean of five seeds). Rows are training corpora, columns test corpora; the diagonal is in-domain. Blue is above chance, red below; the midpoint is 0.5.

most predictive of *corpus identity* (from a one-vs-rest corpus classifier); and three dense variants on a 300-d SVD of the TF-IDF space: the SVD baseline, *INLP* (Ravfogel et al., 2020), which projects out 15 iterations of linear corpus-identity directions, and *CORAL* (Sun et al., 2016), which aligns source feature covariance to the (unlabelled) target. None uses target labels; CORAL alone uses unlabelled target text.

5. Results

5.1. The size of the gap

Table 4 and Figure 1 summarise the seven-corpus veracity matrix. With TFIDF-LR, mean in-domain ROC-AUC is 0.914; mean off-diagonal AUC is 0.567, and on the eight same-task cells 0.624. The sanity controls behave as they should: with permuted labels, transfer AUC is 0.502 ($\pm$0.067 across cells and seeds) and in-domain AUC 0.514, and with every source trained on 250 items the gap remains (in-domain 0.861, off-diagonal 0.556), so it is not produced by the very different corpus sizes. The gap is therefore real, but it is not a collapse to chance: on the cleanest pair, ISOT$\rightarrow$WELFake, AUC is 0.69 [0.66, 0.71], and in the reverse direction 0.85 [0.82, 0.87] (Table 5). Two pairs carry almost no transferable ranking: PolitiFact$\rightarrow$LIAR (0.50 [0.47, 0.54]) and LIAR$\rightarrow$PolitiFact (0.51 [0.42, 0.60]), even though both are labelled by the same fact-checker, because one rates claims and the other news stories. 8 of the 42 off-diagonal cells (19%) have a mean AUC below 0.5: the cue a source teaches is systematically *reversed* on that target.

Table 4: In-domain versus transfer performance on the seven veracity corpora (mean ± s.d. over five seeds of the per-seed mean over cells). "Pairs" averages the eight directed same-task cells. Off-diagonal macro-F1 is given at the default threshold and at the oracle threshold. ROBERTA-LR is evaluated on the 1,500-item subsets (Section 4.2).

		ROC-AUC			macro-F1		
Model	Regime	In-domain	Off-diag.	Pairs	In-domain	Off-diag.	Off-diag. oracle
TFIDF-LR	full	0.914 ±0.006	0.567 ±0.006	0.624 ±0.004	0.854 ±0.005	0.439 ±0.011	0.565 ±0.004
TFIDF-LR	matched	0.861 ±0.010	0.556 ±0.004	0.590 ±0.009	0.786 ±0.012	0.446 ±0.014	0.558 ±0.002
TFIDF-LR	permuted	0.519 ±0.026	0.499 ±0.012	0.514 ±0.019	0.397 ±0.014	0.377 ±0.015	0.518 ±0.007
TFIDF-SVM	full	0.914 ±0.005	0.564 ±0.005	0.625 ±0.005	0.855 ±0.005	0.437 ±0.009	0.564 ±0.004
TFIDF-SVM	matched	0.862 ±0.010	0.554 ±0.005	0.592 ±0.010	0.787 ±0.011	0.447 ±0.017	0.556 ±0.003
TFIDF-SVM	permuted	0.519 ±0.024	0.498 ±0.015	0.511 ±0.020	0.398 ±0.009	0.377 ±0.012	0.516 ±0.008
NB-SVM	full	0.912 ±0.004	0.547 ±0.006	0.590 ±0.011	0.853 ±0.008	0.410 ±0.008	0.555 ±0.003
NB-SVM	matched	0.862 ±0.011	0.540 ±0.006	0.559 ±0.004	0.786 ±0.012	0.398 ±0.013	0.549 ±0.004
NB-SVM	permuted	0.510 ±0.023	0.494 ±0.009	0.502 ±0.009	0.400 ±0.003	0.383 ±0.012	0.520 ±0.006
GLOVE-LR	full	0.864 ±0.005	0.564 ±0.005	0.610 ±0.005	0.782 ±0.007	0.465 ±0.004	0.564 ±0.004
GLOVE-LR	matched	0.830 ±0.006	0.563 ±0.010	0.592 ±0.017	0.740 ±0.011	0.466 ±0.010	0.564 ±0.007
GLOVE-LR	permuted	0.520 ±0.022	0.504 ±0.026	0.494 ±0.021	0.404 ±0.020	0.391 ±0.012	0.525 ±0.019
STYLE-GB	full	0.745 ±0.009	0.535 ±0.004	0.595 ±0.008	0.679 ±0.005	0.427 ±0.007	0.548 ±0.003
STYLE-GB	matched	0.702 ±0.008	0.526 ±0.009	0.561 ±0.032	0.632 ±0.008	0.413 ±0.017	0.542 ±0.007
STYLE-GB	permuted	0.503 ±0.018	0.502 ±0.016	0.493 ±0.028	0.410 ±0.008	0.390 ±0.007	0.523 ±0.011
ROBERTA-LR	full	0.861 ±0.012	0.544 ±0.008	0.630 ±0.011	0.777 ±0.013	0.452 ±0.013	0.549 ±0.010

Threshold-dependent metrics tell a harsher story than AUC. Off-diagonal macro-F1 at the default threshold averages 0.439, barely above the one-class level of 0.33, against 0.854 in-domain. The reason is visible in the predicted-positive rates (PPR, Table 5): an ISOT model flags 0.78 of WELFake as fake although the fake rate is 0.43, and a GossipCop model flags only 0.07 of PolitiFact although its fake rate is 0.43. Scores that were well calibrated in-domain are badly calibrated on the target, which brings us to the decomposition.

5.2. Where the gap goes

Applying Eq. (1) cell by cell (Table 6, Figure 2) splits the loss into a part that no threshold can recover and a part that the right threshold would. For TFIDF-LR on the same-task pairs, the mean transfer gap is 0.31 macro-F1 points, of which 0.24 (79%) is a ranking deficit and the rest an operating-point loss. On the remaining cells the gap is 0.44, with 68% attributable to ranking. Across all cells, 70% of the loss is ranking and 30% operating point. The gap is therefore *mostly ranking, partly threshold.* Both parts matter: the

Table 5: Same-task pairs (TFIDF-LR, five seeds). AUC with pooled seed-bootstrap 95% CI; "In" is the source's in-domain AUC; prior is the target fake rate and PPR the predicted-positive rate at $t = 0.5$; macro-F1 at the default, prior-matched and oracle thresholds; ECE of the source-calibrated scores on the target.

Pair	In	AUC [95% CI]	Prior	PPR	$F1_{0.5}$	$F1_{\pi}$	$F1^{\star}$	ECE
ISOT→WELFake	0.999	0.690 [0.66, 0.71]	0.43	0.78	0.539	0.635	0.638	0.38
WELFake→ISOT	0.989	0.847 [0.82, 0.87]	0.46	0.36	0.753	0.763	0.765	0.11
PolitiFact→GossipCop	0.893	0.526 [0.48, 0.56]	0.23	0.54	0.464	0.514	0.517	0.32
GossipCop→PolitiFact	0.869	0.589 [0.50, 0.68]	0.43	0.07	0.425	0.576	0.595	0.28
Twitter15→Twitter16	0.981	0.672 [0.52, 0.82]	0.46	0.62	0.572	0.597	0.646	0.25
Twitter16→Twitter15	0.967	0.657 [0.55, 0.75]	0.51	0.63	0.580	0.591	0.631	0.28
LIAR→PolitiFact	0.698	0.508 [0.42, 0.60]	0.44	0.54	0.495	0.517	0.536	0.12
PolitiFact→LIAR	0.893	0.505 [0.47, 0.54]	0.44	0.40	0.497	0.502	0.509	0.22

operating-point share is large enough to change which of two detectors looks better, and it is the cheaper one to fix.

5.3. Repairing the operating point without labels

Table 7 and Figure 3 compare operating points. The textbook remedy, EM prior adaptation, *lowers* off-diagonal macro-F1 from 0.439 to 0.358, i.e. it recovers −64% of the operating-point loss. Rank transfer, which keeps the source's base rate and simply flags the same fraction of target items, raises macro-F1 to 0.541 and recovers 81% of the loss, close to the 86% achieved when the true target base rate is known (0.547); the oracle threshold reaches 0.565. A 2-component mixture recovers 57%, and flagging half of the items 80%.

The reason EM fails is diagnostic. EM is optimal under label shift, where only $p(y)$ changes and calibrated posteriors stay meaningful. Here calibration does not survive the move: expected calibration error is 0.046 in-domain but 0.328 on the target, so EM's prior estimates drift towards 0 or 1 (mean absolute error 0.46) while the source prior is a far better guess of the target prior (error 0.09). The shift is in $p(x \mid y)$ (style, topic and source), not in $p(y)$, and the right label-free response is to transfer the ranking-based decision rule rather than the score scale.

5.4. The gap across models

The picture does not depend on the detector (Table 4; full matrices for every model are in the Supplementary Material). The four content models reach in-domain AUC between 0.864 and 0.914 and off-diagonal AUC between 0.547 and 0.567, and all of them show the same split between ranking and operating-point loss (Table 6). Averaged word embeddings (GLOVE-LR) are weaker in-domain (0.864) without transferring better (0.564).

Table 6: Transfer gap decomposition (Eq. 1), macro-F1 points, mean over cells and seeds. The gap is relative to the target's own in-domain score.

Model	Cells	Gap	Ranking	Threshold	Prior/cal.	Ranking share
TFIDF-LR	pairs	0.308	0.244	0.018	0.046	79%
TFIDF-LR	other cells	0.441	0.300	0.018	0.122	68%
TFIDF-SVM	pairs	0.308	0.243	0.016	0.049	79%
TFIDF-SVM	other cells	0.444	0.303	0.018	0.123	68%
NB-SVM	pairs	0.319	0.261	0.023	0.036	82%
NB-SVM	other cells	0.472	0.306	0.021	0.145	65%
GLOVE-LR	pairs	0.281	0.193	0.016	0.071	69%
GLOVE-LR	other cells	0.327	0.224	0.017	0.085	69%
STYLE-GB	pairs	0.141	0.091	0.013	0.036	65%
STYLE-GB	other cells	0.278	0.141	0.020	0.117	51%
ROBERTA-LR	pairs	0.243	0.162	0.022	0.059	66%
ROBERTA-LR	other cells	0.344	0.244	0.024	0.076	71%

The style model is the most telling. Seeing only 22 surface statistics, it reaches in-domain AUC 0.745 on average (0.999 on ISOT, 0.951 on WELFake), and its transfer AUC (0.535) is only a few points below that of the lexical models, so a large part of what the benchmarks reward is form: length, casing, punctuation and attribution habits (Section 5.5).

A pretrained encoder does not change the conclusion. On identical 1,500-item subsets, RoBERTa representations give in-domain AUC 0.861 and off-diagonal AUC 0.544, against 0.895 and 0.549 for TFIDF-LR refitted on the same splits; the paired off-diagonal difference is -0.005 AUC (Wilcoxon $p = 0.44$), and RoBERTa is better in 17 of the 42 cells. This agrees with Verma (2026) and Dell'Oglio et al. (2026), who found that fine-tuned transformers gain in-domain but not, or not reliably, across corpora. Pairwise McNemar tests on the same target items (Holm-corrected over all comparisons) find significant differences from TFIDF-LR in the following share of (cell, seed) comparisons: GLOVE-LR 36%, NB-SVM 21%, ROBERTA-LR 17%, STYLE-GB 47%, TFIDF-SVM 12%.

5.5. What the detectors learn

Cheap signals go a long way. Table 8 lists four shortcut indicators per corpus. The surface-statistics model recovers on average 57% of the above-chance in-domain AUC of the full lexical model, and on the article corpora almost all of it (ISOT 0.999, WELFake 0.951). Sixty-four training examples are enough for an in-domain AUC of 0.983 on ISOT and 0.894 on WELFake; the claim and celebrity-headline corpora, whose in-domain scores are lower to begin with, learn much more slowly (LIAR 0.567, GossipCop 0.610). On

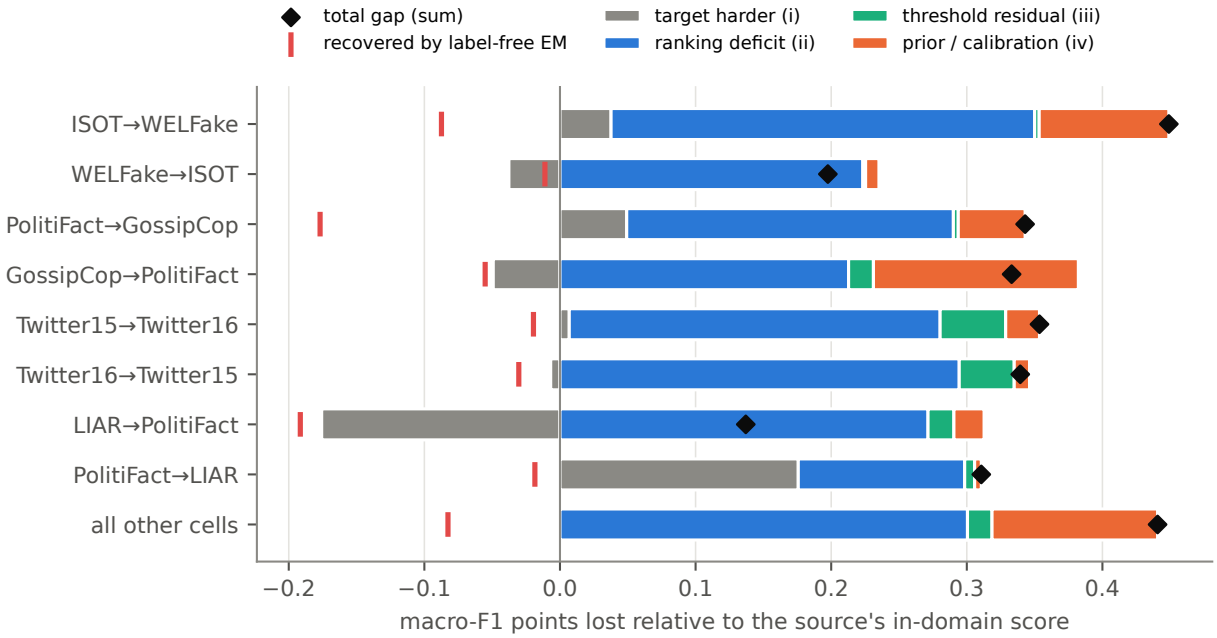


Figure 2: Transfer gap decomposition for TFIDF-LR (macro-F1 points relative to the source's in-domain score, mean of five seeds). Grey: target harder or easier than source; blue: ranking deficit; green and orange: operating-point losses. Diamonds mark the total; red ticks show what EM prior adaptation recovers (negative values make things worse).

the headline and claim corpora, publisher or speaker identity alone predicts the label with AUC 0.840 (PolitiFact), 0.807 (GossipCop) and 0.664 (LIAR). A label that can be predicted from form, from a few dozen examples or from who published an item is a label whose in-domain score says little about veracity.

The strongest cues are collection signatures. Figure 4 shows the highest-weight n-grams. In ISOT, the real class is signalled by Reuters wire conventions ("reuters", "said", "washington reuters", weekday attributions such as "on thursday") and the fake class by the template of the sites it was collected from ("featured image", "image via", "read more", "video"). In WELFake, the real class is signalled by publisher suffixes inherited from one of its source collections ("breitbart", "york times", "mr") and the fake class by the window in which fake items were gathered ("2016", "october", "november", "hillary"). Each corpus thus teaches a different, corpus-specific proxy, which explains why the sign of a cue can flip from one corpus to another (Section 5.1).

Removing cues does not remove the signal. Deleting the top-K features and retraining barely moves in-domain AUC: on ISOT it goes from 0.999 with all features to 0.991 after deleting 1,000 and 0.985 after 3,000, and on WELFake from 0.984 to 0.962 and 0.949; transfer to the paired corpus, in contrast, falls from 0.673 to 0.580 (ISOT→WELFake) and from 0.827 to 0.718 (WELFake→ISOT). The highest-weight features thus carry most of what transfers, while the signal that keeps in-domain AUC near its ceiling is spread over thousands of corpus-specific features. Masking source tags, datelines, time expressions, num-

Table 7: Off-diagonal macro-F1 under different operating points (mean over the 42 cells and five seeds). Label-free: default $t = 0.5$, EM prior adaptation, rank transfer (predicted-positive rate set to the source prior), predicted-positive rate 0.5, and a 2-component GMM on target logits. With one number known: rate set to the true target prior. Upper bound: oracle threshold. For TFIDF-LR, Holm-corrected Wilcoxon p versus default: EM 1.6e-32, rank transfer 6.6e-31, rate 0.5 1.4e-29, GMM 1.7e-18.

	label-free					prior known	bound
Model	$t = 0.5$	EM	rank transfer	rate 0.5	GMM	rate $= \pi_B$	oracle
TFIDF-LR	0.439	0.358	0.541	0.539	0.510	0.547	0.565
TFIDF-SVM	0.437	0.363	0.539	0.537	0.503	0.546	0.564
NB-SVM	0.410	0.349	0.525	0.526	0.444	0.534	0.555
GLOVE-LR	0.465	0.381	0.540	0.542	0.494	0.547	0.564
STYLE-GB	0.427	0.347	0.521	0.521	0.477	0.529	0.548
ROBERTA-LR	0.452	0.368	0.516	0.516	0.477	0.525	0.549

bers and named entities together (Table 9) leaves in-domain AUC at 0.996 (ISOT) and 0.964 (WELFake), and moves ISOT→WELFake from 0.679 to 0.650 and WELFake→ISOT from 0.808 to 0.810. Balancing classes within each of 50 topic clusters, whose class purity averages 0.82 in ISOT and 0.81 in WELFake, leaves in-domain AUC unchanged (ISOT 0.998→0.999, WELFake 0.984→0.986) and transfer unchanged (0.604→0.606, 0.622→0.622); only GossipCop loses in-domain AUC (0.846→0.799). The separating signal is therefore diffuse: it lives in style and register throughout the text, not in a removable list of tokens or topics, as Verma (2026) also found for ISOT.

Corpus divergence does not predict transfer. Every pair of corpora can be told apart from text alone (proxy $\mathcal{A}$-distance between 1.28 and 1.98, maximum 2; ISOT–WELFake 1.45), yet across the 42 cells we find no association between source–target divergence and transfer AUC (Spearman $\rho = -0.01$). With seven corpora the corpus-bootstrap 95% interval is wide ([-0.61, 0.69]), so a moderate association cannot be excluded, but label-free divergence is clearly not a usable predictor of transfer here. Transfer depends less on how different two corpora look than on whether the cue that separates the classes in one also separates them, with the same sign, in the other.

5.6. Repairing the ranking

Table 10 reports the leave-one-corpus-out repairs (per-target values in Table 11). Training on the six other corpora at once is the one change that helps ranking: held-out AUC rises to 0.660, against 0.567 for the average single source. This is close to the best single source chosen with hindsight (0.671), which cannot be identified without target labels, and far below the targets' own in-domain AUC of 0.914. Nothing we add on top of pooling helps on average. Balancing every (corpus, class) cell (0.659, Δ-0.001, $p = 1.000$) and

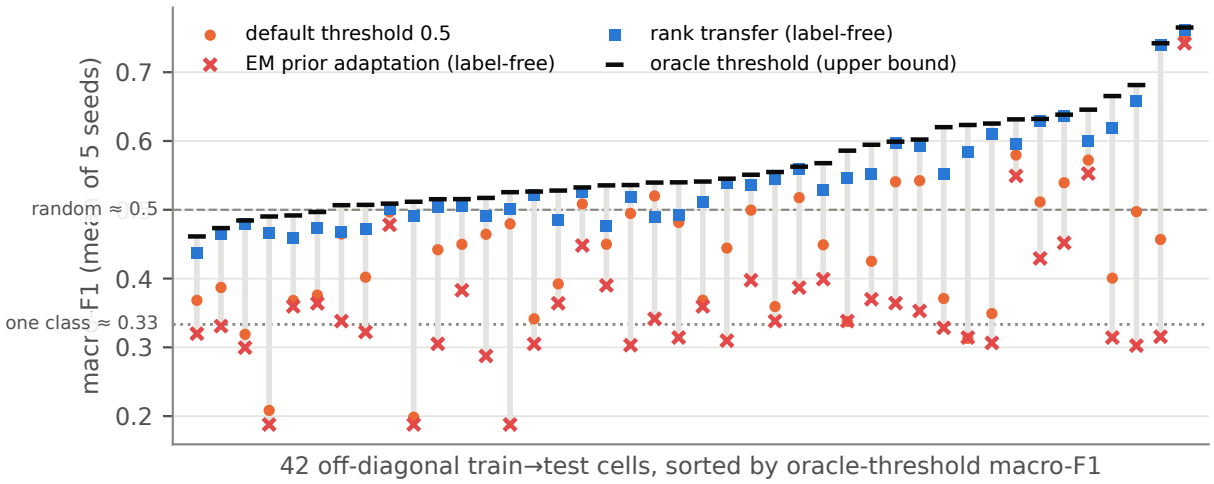


Figure 3: Off-diagonal macro-F1 of TFIDF-LR for each of the 42 cells (mean of five seeds), sorted by the oracle bound. Rank transfer closes most of the distance from the default threshold to the oracle without any target label; EM prior adaptation often falls below the default.

masking collection signatures (0.661, Δ+0.001, $p = 1.000$) leave AUC unchanged on average; masking helps clearly only on ISOT (Table 11), the corpus whose signatures it was written for.

Methods that make the representation *invariant to corpus identity* make ranking worse: deleting the 2,000 most corpus-predictive features gives 0.636 (Δ-0.023, $p = 0.006$), CORAL 0.621 (Δ-0.038, $p =< 0.001$) and INLP 0.605 (Δ-0.054, $p = 0.001$). INLP does what it is designed to do, reducing linear corpus-identity accuracy from 0.87 to 0.25 (six classes), but it removes veracity information with it. In these benchmarks corpus identity and label are entangled: each corpus collected its fake and real items from different outlets, periods and formats, so the directions that identify a corpus are partly the directions that separate its classes. Invariance to the corpus is therefore not invariance to the shortcut, and the remaining ranking gap is a property of the data rather than of the learner.

5.7. A low-resource language: Bengali clickbait

Replication and duplicates. On the official BanglaBait split, character 3–5-gram logistic regression reaches a clickbait-class F1 of 64.37 (Table 12), within two points of the 66.28 reported for the same model by Mahtab et al. (2023), and a linear SVM reaches 63.79. Removing the near-duplicates found by MinHash from the official splits lowers the same model to 59.15 (AUC 0.834→0.823): even after the original authors' Levenshtein filter, duplicated titles inflate the benchmark. Over five random re-splits of the deduplicated corpus the model reaches AUC 0.821 ± 0.011.

Publisher identity. The 19 publishers with at least 100 articles have clickbait rates from 7% to 66%, so publisher identity alone predicts the label with AUC 0.687 (Section 5.5). Leave-one-publisher-out evaluation

Table 8: Shortcut indicators (ROC-AUC, mean of five seeds). In-domain AUC of TFIDF-LR with the full training split, of the surface-statistics STYLE-GB, and of TFIDF-LR trained on only 64 or 256 items; AUC of a group-identity baseline (publisher domain for PolitiFact and GossipCop, speaker for LIAR); and the mean off-diagonal AUC with the corpus as source and as target.

	in-domain					transfer	
Corpus	full	style only	$n = 64$	$n = 256$	group only	as source	as target
ISOT	0.999	0.999	0.983	0.991	–	0.611	0.670
WELFake	0.989	0.951	0.894	0.941	–	0.626	0.540
PolitiFact	0.893	0.787	0.742	0.850	0.840	0.533	0.592
GossipCop	0.869	0.659	0.610	0.723	0.807	0.538	0.522
LIAR	0.698	0.590	0.567	0.607	0.664	0.537	0.530
Twitter15	0.981	0.603	0.793	0.962	–	0.572	0.563
Twitter16	0.967	0.625	0.890	–	–	0.556	0.552

nevertheless costs little: across 20 held-out publishers (the smallest ones pooled), mean AUC is 0.791 (range 0.651–0.911), close to the random-split value. Bengali clickbait detection, in other words, survives a change of source that English veracity detection does not. We read this as a construct effect rather than a language effect: clickbait *is* a property of a headline's style, so a style-sensitive detector learns the construct itself, whereas for veracity style is only a correlate of the label that changes from corpus to corpus.

Across languages. A translate-test probe uses BanglaBait's machine-translated English titles. A detector trained on English clickbait headlines (Chakraborty et al., 2016) ranks translated Bengali titles with AUC 0.707, against 0.777 for a detector trained on the translated titles themselves and 0.813 for one trained on the original Bengali; translation costs about 0.04 AUC and most of the in-language ranking ability carries over. In the other direction, the translated-Bengali detector reaches 0.931 on English headlines (in-domain 0.994). Clickbait transfers across languages far better than veracity transfers across English corpora.

6. Discussion

What the benchmarks measure. Every corpus in our suite can be told apart from every other with high accuracy (proxy $\mathcal{A}$-distance 1.28–1.98 on a scale whose maximum is 2); on the article corpora nearly all of the in-domain signal, and on average more than half of it, is available to a model of surface statistics; the article and tweet corpora reach most of their final AUC with 64 training examples; and the heaviest feature weights point to outlet signatures and to the time window in which the fake class was collected (Section 5.5). In-domain scores therefore measure, to a large extent, how well a model recognises the *collection process* that produced each class. This is the same conclusion Verma (2026) reached for ISOT alone; our matrix

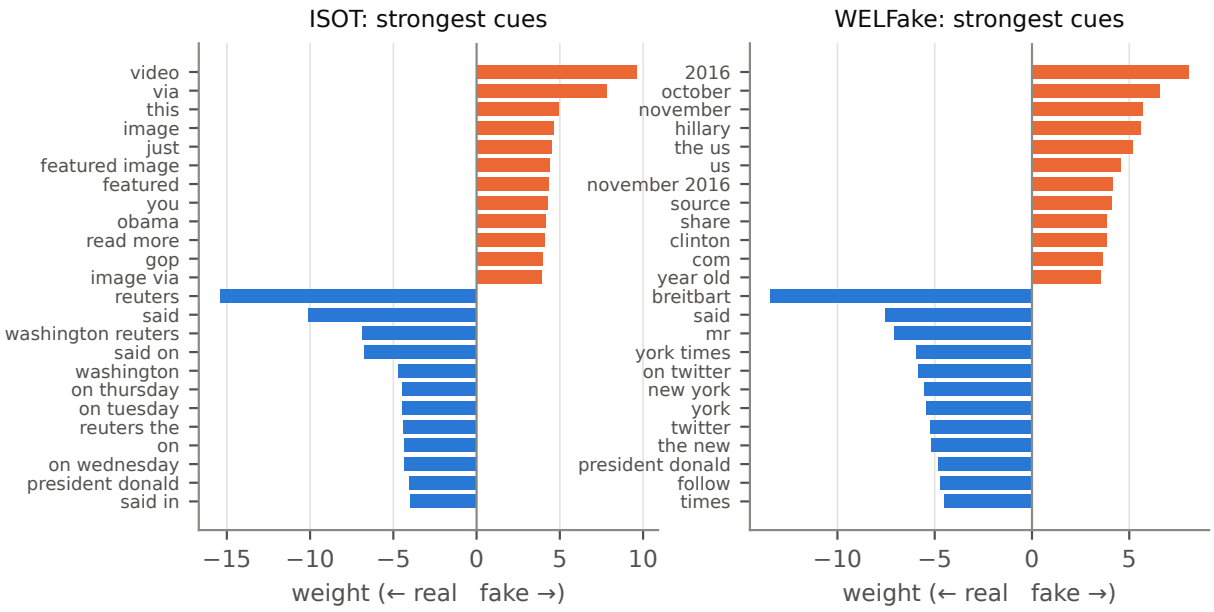


Figure 4: Highest-weight n-grams of TFIDF-LR (seed 0) for the real (blue) and fake (orange) classes. Both corpora are separated mainly by outlet conventions and collection-period vocabulary.

shows that it holds across units and constructs, and that the resulting loss in a new corpus is mostly a loss of ranking, not of calibration.

What to do about it. The decomposition separates a cheap problem from a hard one. The operating-point part is cheap: rank transfer needs no target labels, is one line of code and recovered 81% of that loss, whereas the textbook label-shift correction made it worse. Practitioners who deploy a detector on a new feed should therefore fix the *fraction* of items they flag, not the score threshold, unless they have reason to believe the base rate has changed and have labelled data to estimate it. The ranking part is hard. Source diversity helped (pooled training reached 0.660 held-out AUC against 0.567 for a single source), but the best label-free invariance methods made ranking worse, and the pooled model is still far below the in-domain ceiling of 0.914. Representation-level invariance to corpus identity is therefore not enough when the label itself is tied to how each corpus was assembled; better ranking needs better data, in particular corpora whose real and fake items come from the same outlets and time windows, or evidence-based detectors that do not rely on style at all (Pelrine et al., 2023; Thibault et al., 2025).

Style as construct and as shortcut. The Bengali results add a distinction that the English matrix alone would not reveal. Clickbait detection survived held-out publishers and even a change of language, while veracity detection did not survive a change of corpus. A style-sensitive model transfers when style *is* the construct, as for clickbait headlines, and fails when style is only correlated with the label inside one collection, as for

Table 9: Masking ablation (TFIDF-LR, ROC-AUC, three seeds, first 3,000 characters). Rows: training corpus and what is masked in every corpus; columns: test corpus (the training corpus's own column is in-domain).

Train	Masked	ISOT	WELFake	PolitiFact	GossipCop	LIAR
ISOT	none	0.998	0.679	0.739	0.481	0.559
ISOT	source tags, datelines, URLs, handles	0.997	0.681	0.733	0.483	0.559
ISOT	dates, weekdays, numbers	0.998	0.692	0.723	0.479	0.563
ISOT	named entities	0.996	0.657	0.678	0.543	0.557
ISOT	all of the above	0.996	0.650	0.678	0.540	0.556
WELFake	none	0.808	0.980	0.636	0.529	0.532
WELFake	source tags, datelines, URLs, handles	0.815	0.979	0.634	0.533	0.536
WELFake	dates, weekdays, numbers	0.831	0.977	0.608	0.509	0.526
WELFake	named entities	0.833	0.965	0.614	0.511	0.522
WELFake	all of the above	0.810	0.964	0.611	0.500	0.522

fake news. Whether a misinformation construct can be learned from text alone is thus construct-specific, and benchmark designers should say which kind of cue their labels license.

Reporting protocol. For any claim that a misinformation detector generalises: (1) *verify label direction* with signature probes and treat a same-task AUC below 0.5 as a data error until shown otherwise; (2) *deduplicate* within and across corpora with near-duplicate matching, and drop label-conflicting groups; (3) report at least one *same-task cross-corpus* result, with a ranking metric (ROC-AUC or AP) next to F1; (4) report the *decomposition*: F1 at the default, prior-matched and oracle thresholds; (5) run three cheap *shortcut probes*: a surface-statistics model, a group-identity baseline where metadata exist, and a 64-example learning-curve point; (6) report a *permutation control*, at least five seeds and bootstrap intervals; (7) deploy with a *rank-based* operating point unless target labels are available.

6.1. Limitations and threats to validity

Models. Our detectors are linear or tree-based on sparse, static-embedding and frozen RoBERTa features; we did not fine-tune transformers end to end or prompt large language models, because the study was run on a two-core CPU. Prior work found fine-tuning to increase in-domain scores without improving transfer (Verma, 2026; Dell'Oglio et al., 2026) and zero-shot language models to be more robust (Pelrine et al., 2023; Dell'Oglio et al., 2026); the decomposition applies unchanged to any scored detector, and testing it on those families is the most direct extension. RoBERTa results use 1,500-item subsets and the first 600 characters of each text. *Corpora.* Labels differ in kind (claims, stories, outlets), which is why we lean on same-task pairs; FakeNewsNet is used through its public titles only; NELA-GT and BanFakeNews could not

Table 10: Leave-one-corpus-out repairs (TFIDF-LR features; mean over the seven held-out veracity corpora and five seeds; s.d. across seeds). ΔAUC versus pooled training with Holm-corrected Wilcoxon p over the 35 (target, seed) pairs. For reference, the mean single-source transfer AUC to the same targets is 0.567, the best single source (chosen with target labels) 0.671, and the targets' own in-domain AUC 0.914.

Training	AUC	AP	$F1_{0.5}$	$F1^{\star}$	ΔAUC (p)
pooled (6 corpora)	0.660 ±0.013	0.582	0.581	0.634	–
+ (corpus, class) balancing	0.659 ±0.013	0.583	0.582	0.632	-0.001 (1.000)
+ balancing + masking	0.661 ±0.012	0.584	0.576	0.633	+0.001 (1.000)
+ balancing + corpus-feature deletion	0.636 ±0.011	0.556	0.560	0.611	-0.023 (0.006)
SVD-300 + balancing	0.647 ±0.009	0.566	0.583	0.624	-0.013 (0.043)
SVD-300 + INLP	0.605 ±0.011	0.520	0.525	0.591	-0.054 (0.001)
SVD-300 + CORAL	0.621 ±0.008	0.539	0.563	0.604	-0.038 (<0.001)

Table 11: Leave-one-corpus-out AUC per held-out corpus (mean of five seeds), against single-source transfer and the in-domain ceiling.

Held out	mean single	best single	pooled	+ masking	INLP	CORAL	in-domain
ISOT	0.670	0.847	0.865	0.890	0.805	0.781	0.999
WELFake	0.540	0.690	0.648	0.657	0.643	0.643	0.989
PolitiFact	0.592	0.730	0.677	0.655	0.600	0.650	0.893
GossipCop	0.522	0.544	0.538	0.546	0.558	0.520	0.869
LIAR	0.530	0.552	0.528	0.528	0.502	0.514	0.698
Twitter15	0.563	0.657	0.647	0.645	0.592	0.626	0.981
Twitter16	0.552	0.678	0.715	0.703	0.536	0.616	0.967

be obtained in our environment, so the Bengali analysis uses one Bengali corpus and a translate-test rather than a second native corpus. PHEME labels rumours, not falsehood, and is used only as an additional unit. Deduplication thresholds (Jaccard ≥ 0.8) trade recall for precision; looser thresholds would remove more items. *Statistics.* Twitter15/16 and PolitiFact test sets are small, so their cells have wide intervals (Table 5); we report pooled seed-bootstrap intervals and avoid conclusions from single small cells. The 42 cells are not independent because they share corpora; where we test across cells we resample corpora, not cells. *Scope.* Our results concern text-only detection. Propagation-based detectors face related spurious-correlation problems on Twitter15/16 and PHEME (Wu and Hooi, 2022; Mu et al., 2023; Pelrine et al., 2021), which we did not re-evaluate. *Ethics.* We use only released text of public posts, no user identifiers and no re-collection; tweet content remains subject to the platform's terms, so we release code, per-cell results and saved scores, but no post text.

Table 12: BanglaBait, official split. Clickbait-class F1, precision and recall and accuracy (%), as reported by Mahtab et al. (2023), plus ROC-AUC. Reference values from that paper (as released): LR char 3–5-grams F1 66.28, BanglaBERT 71.72, SS-GAN BanglaBERT 75.13.

Test set	Input	Model	F1	P	R	Acc.	AUC
as released	title	LR, char 3–5	64.37	73.63	57.17	77.72	0.834
as released	title	SVM, char 3–5	63.79	73.37	56.42	77.46	0.836
as released	title	LR, word 1–2	48.51	65.59	38.49	71.25	0.737
as released	title+content	LR, char 3–5	61.29	71.92	53.40	76.26	0.792
as released	title+content	SVM, char 3–5	60.68	71.78	52.55	76.03	0.791
as released	title+content	LR, word 1–2	57.48	70.36	48.58	74.70	0.775
deduplicated	title	LR, char 3–5	59.15	70.72	50.84	76.57	0.823
deduplicated	title	SVM, char 3–5	60.01	71.23	51.85	76.94	0.824
deduplicated	title	LR, word 1–2	40.77	64.56	29.79	71.11	0.722
deduplicated	title+content	LR, char 3–5	52.34	67.98	42.55	74.14	0.772
deduplicated	title+content	SVM, char 3–5	53.28	69.42	43.23	74.70	0.772
deduplicated	title+content	LR, word 1–2	48.61	66.86	38.19	73.06	0.752

7. Conclusion

Misinformation detectors that look near-perfect in-domain lose much of their value on a new corpus, but not all of it, and not for one reason. Across ten audited corpora, six detectors and five seeds, 70% of the macro-F1 they lose is a ranking deficit that only better data or representations can address, and 30% is a miss-set operating point that can be repaired without a single target label, provided one transfers the detector's ranking rule rather than trusting its scores or a label-shift correction. Much of what in-domain benchmarks reward is the collection process behind each class: outlet signatures, attribution style, time windows. Where style is the construct itself, as for clickbait, detection travels across publishers and languages; where it is a by-product of how a corpus was assembled, as for veracity, it does not. Benchmarks and papers should report transfer, ranking metrics and the decomposition alongside in-domain accuracy, and the protocol and code we release make that cheap to do.

CRediT authorship contribution statement

Dewan Nafay Chowdhury: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Md. Rejaul Korim Sadi:** Conceptualization, Methodology, Validation, Investigation, Data curation, Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All corpora are publicly available from their original releases (Section 3 and the repository's `DATA.md`); none is redistributed. Code, fixed seeds, per-cell results, saved predictions, logs and a one-command reproduction script are released under the MIT licence at `https://github.com/<account>/fn-transfer-audit` (archived release DOI to be added on acceptance).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with code development, literature search and drafting of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Implementation details and compute

Hyperparameters. Logistic regression and linear SVM: $C \in \{0.25, 1, 4\}$ (SVM: $C/4$), liblinear, selected by validation ROC-AUC; TF-IDF word 1–2-grams, `min_df`=2, 100,000 features, sublinear term frequency, texts truncated to 20,000 characters. NB-SVM: binary 1–2-grams, add-one smoothed log-count ratios. GloVe: 100-d `glove-wiki-gigaword`, mean of in-vocabulary tokens. Style model: histogram gradient boosting with 100 or 300 iterations, learning rate 0.1. RoBERTa: mean of last-layer piece states, standardised, logistic regression with $C \in \{0.025, 0.1, 0.4\}$. Platt scaling on the validation split for every model. Bengali: character 3–5-grams (`char_wb`), 200,000 features.

Deduplication. MinHash with 128 permutations (seed 1) and LSH at Jaccard 0.8; word 3-gram shingles over the first 400 words for articles, character 5-grams otherwise; text normalised with NFKC, lower-casing, URL and punctuation removal.

Compute. All experiments ran on a 2-core CPU with 8 GB RAM and no GPU; the full pipeline (`run_all.sh`) takes about ten to twelve hours. Software: Python 3.13, scikit-learn 1.9.1, spaCy 3.8, PyTorch 2 (CPU), datasketch, statsmodels.

Appendix B. Reproducibility checklist

- Data: public sources, licences and expected layout in `DATA.md`; label harmonisation and its verification in `src/label_probe.py`; deduplication counts in `results/dedup_report.json`.
- Splits and seeds: five seeds (0–4) fix splits, subsampling and model initialisation; vectorisers and calibrators are fitted on source training and validation data only.
- Every number in this paper is written by `src/paper_numbers.py` from the files in `results/`; no number is typed by hand.
- Saved per-item scores for every (model, source, target, seed) cell allow any metric or threshold rule to be recomputed without retraining.
- Sanity controls: permutation regime (AUC ≈ 0.5) and size-matched regime are part of the main grid.
- The Supplementary Material contains the full matrices for every model, per-cell decompositions and operating points, learning curves, the vocabulary ablation, per-publisher Bengali results and the deduplication report.